MAILTRACE AI // FORENSIC TELEMETRY PLATFORM

# Digital Evidence & Forensic Incident Dossier

| | | | |
|---|---|---|---|
| **REPORT REFERENCE:** | RPT-■20260910-■77E97B | **GENERATED (UTC):** | 2026-■09-■10 18:■43:■02 UTC |
| **EVIDENCE IDENTIFIER:** | EVD-■5E58C282C0 | **ASSIGNED CASE:** | UNASSIGNED /■ DIRECT EVIDENCE |
| **INVESTIGATOR:** | Lead Forensic Examiner | **CLASSIFICATION:** | **CONFIDENTIAL // EVIDENTIARY DISCLOSURE** |
| **EMAIL SHA-256:** | `5e58c282c0ac76456bf139f43214■0e0dfc5338ca13502c9bea990d35■aa825c3b` | **HASH ALGORITHM:** | FIPS PUB 180-4 SHA-256 (Preserved Literal) |

## 1. Executive Forensic Summary

| DETERMINISTIC THREAT SCORE | ASSESSED SEVERITY | ML PHISHING PROBABILITY | ML CLASSIFICATION |
|---|---|---|---|
| **38/100** | **SUSPICIOUS** | **65.5%** | **PHISHING** |

Email risk assessed as Suspicious (38/100) driven by 3 forensic signal(s): Possible PayPal lookalike domain (+18), Suspicious URL detected (+8), Credential harvesting or urgent security language (+12).

## 2. Email Headers & Sender Alignment

| | |
|---|---|
| **From:** | `security-■alert@■paypal-■update.■com` |
| **To:** | `victim@■company.■com` |
| **Subject:** | **Urgent Security Alert:■ Account Suspicious Access** |
| **Date:** | Tue,■ 08 Sep 2026 10:■30:■00 +■0000 |
| **Reply-To:** | `None` |
| **Return-Path:** | `None` |
| **Message-ID:** | `&■lt;■evd-■test-■12345@■paypal-■update.■com&■gt;■` |

| | | | |
|---|---|---|---|
| **From Domain:** | `paypal-■update.■com` | **Reply-To Domain:** | N/■A **(Aligned)** |
| **Return-Path Domain:** | N/■A **(Aligned)** | **Alignment Verdict:** | **Sender domains aligned** |

## 3. Email Authentication Findings (SPF / DKIM / DMARC)

| PROTOCOL | OBSERVED VERDICT | TECHNICAL DETAILS & EVIDENCE |
|---|---|---|
| **SPF (Sender Policy)** | NONE | No Authentication-■Results or Received-■SPF header present.■ |
| **DKIM (Cryptographic)** | NONE | No Authentication-■Results or DKIM-■Signature header present.■ |
| **DMARC (Domain Policy)** | NONE | No Authentication-■Results header present.■ |

*Notice: Observed authentication result from supplied headers (unverified by local mail server)*

## 4. Deterministic Threat Score Breakdown

| CATEGORY | IMPACT | FORENSIC FINDING & JUSTIFICATION |
|---|---|---|
| **General** | **+18 pts** | N/A |
| **General** | **+8 pts** | N/A |

| | | |
|---|---|---|
| **General** | **+12 pts** | N/A |

## 5. NLP Content Phishing Classification

| | | | |
|---|---|---|---|
| **Model Name:** | TF-■IDF +■ Logistic Regression | **Classification:** | **PHISHING** |
| **Phishing Probability:** | **65.5%** | **Model Confidence:** | medium |
| **Top Text Features:** | `httpurl,■ account,■ immediately,■ security,■ urgent` | **Signal Role:** | Auxiliary ML signal (Non-overriding) |

## 6. AI Analyst Assistant Synthesis

| | | | |
|---|---|---|---|
| **Likely Attack Type:** | **Credential Harvesting Phishing** | **Likely Objective:** | Theft of corporate directory or ■single-■sign-■on credentials via spoofed inter■faces |

**Executive Synthesis:** The analyzed email presents characteristics of Credential Harvesting Phishing with a deterministic threat score of 38/100 (SUSPICIOUS). Key driving indicators include 3 forensic flag(s). Adversary objective is assessed as: Theft of corporate directory or single-sign-on credentials via spoofed interfaces.

**Key Forensic Evidence Identified by LLM:**
- Global threat score evaluated at 38/100 (SUSPICIOUS severity).
- Forensic indicator: Possible PayPal lookalike domain
- Forensic indicator: Suspicious URL detected
- Forensic indicator: Credential harvesting or urgent security language
- Brand lookalike detected: 'paypal-update.com' targeting 'PayPal'.

## 7. Extracted Indicators of Compromise (IOCs)

### Observed URLs & Suspicion Scoring:

| EXTRACTED URL / TARGET HOST | SCORE | FORENSIC OBSERVATIONS & FLAGS |
|---|---|---|
| `http:■/■/■paypal-■update.■com/■login` | **30/100** | Potential brand impersonation:■ similar to paypal.■com (90%■ match),■ Suspicious security/■authentication keywords found:■ login,■ update |

### Observed Attachments & Cryptographic Checksums:

No file attachments detected in this message.

## 8. Network Transmission Path & Topology

| | | | |
|---|---|---|---|
| **Earliest Observable Sending IP:** | **`None identified`** | **Reverse DNS / Host:** | Unknown |

No sequential relay hops reconstructed from Received: headers.

*Notice: Labeled strictly as observed infrastructure route. Does not represent physical attacker travel path.*

## 9. Infrastructure & Domain Intelligence

No public IP intelligence enriched for this email.

## 10. Brand Impersonation & Lookalike Domains

| SUSPECTED BRAND | SPOOFED DOMAIN | SIMILARITY | TECHNIQUES & ATTACK PROFILE |
|---|---|---|---|
| **PayPal** | paypal-■update.■com | **90.0%** | hyphenation,■ brand_■keyword,■ added_■affix |

## 11. Prescriptive Recommendations & Containment Actions

**[ ]** 1. Tag message header with [SUSPICIOUS] warning banner for internal recipient.
**[ ]** 2. Monitor user account for anomalous authentication or link traversal events.

## 12. Evidentiary Limitations & Chain-of-Custody Certification

**Forensic Chain-of-Custody Certification:**
This document is generated by MailTrace AI based on literal RFC-822 evidence headers. Public IP geolocation represents coarse regional estimates based on RIR allocations and does not establish street-level physical locations or the personal identity of the sender. All cryptographic hash sums (SHA-256) were computed over unaltered message byte streams and preserve complete digital non-repudiation chain of custody.

| | | | |
|---|---|---|---|
| **EVIDENTIARY DIGITAL SEAL:** | SHA256:5e58c282c0ac76456bf139f43214■0e0dfc5338ca13502c9bea990d35■aa825c3b | **VERIFICATION STATUS:** | **CRYPTOGRAPHICALLY VERIFIED** |